

Keywords

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Byproducts Rule (D/DBPR)
9210p On-line TOC Analyzer
Fulvic Acid
Humic Acid
Natural Organic Matter (NOM)
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Standard Method SM 5310C

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A Comparative Study of On-line and Laboratory TOC Analyzers for Analysis of Raw and Finished Drinking Water

Introduction

Total organic carbon (TOC) analysis is an important indicator of water quality throughout the drinking water treatment process. Raw source water is progressively treated in chemical coagulation, flocculation, sedimentation, and filtration steps to remove particulate matter and natural organic matter (NOM). Humic acid and fulvic acids contained in the residual NOM of water undergoing disinfection by chlorination react with chlorine to form disinfection by-product compounds such as trihalomethanes (THMs) and haloacetic acids (HAAs), which have been linked to cancer. THMs continue to form during drinking water distribution due to excess chlorine levels required to maintain microbial disinfection.

Disinfectants and Disinfection Byproducts Rule

The U.S. EPA has issued two rules regulating levels of disinfectants and disinfection by-products in drinking water. The Stage 1 Disinfectants and Disinfection Byproducts Rule (D/DBPR) was promulgated in 1998⁽¹⁾ and took effect on January 1, 2002. This rule lowered permissible levels of trihalomethanes (THMs) to 80 µg/L and regulated levels for five haloacetic acids (HAAs), bromate and chlorite in drinking water for the first time.⁽²⁾

The USEPA Stage 2 D/DBPR was promulgated in 2006.⁽³⁾ Compliance dates for the Stage 2 D/DBPR are phased in over time based upon the number of people served by a water system. The compliance date for systems serving over 100,000 people was April 1, 2012. Systems serving 10,000-49,000 people and less than 10,000 people must be in compliance by October 1, 2013.⁽⁴⁾ TOC analysis is an indicator of NOM and THM levels in source water and finished drinking water.⁽⁵⁾

Drinking Water Security

The U.S. Government has passed legislation and issued directives to protect the nations' water supply infrastructure⁽⁶⁾. Title IV of the Public Health Security and Bioterrorism Preparedness and Response Act requires drinking water systems serving more than 3,300 people to develop response measures to incidents that could disrupt the supply of safe drinking water, or pose serious public health concerns⁽⁷⁾. Under Homeland Security Presidential Directive 9 the USEPA is responsible for developing a "robust, comprehensive surveillance and monitoring program to provide early warning in the event of a terrorist attack using biological, chemical, or radiological contaminants"⁽⁸⁾.

On-line TOC monitoring of drinking water supply systems provides warning of organic chemical contamination from accidental or intentional

incidents. TOC is an effective monitoring parameter because it responds to all types of organic carbon dissolved or suspended in water including compounds without a chromophore.

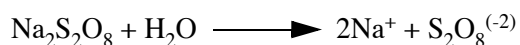
Test methods for regulatory compliance reporting are currently based upon laboratory TOC analysis. On-line TOC analyzers are generally used for process control and need not operate under the quality assurance requirements specified in USEPA methods. Nonetheless, on-line TOC analyzers must be calibrated periodically to ensure accuracy, minimize drift, and verify oxidation efficiency.

This application note presents comparative data obtained on raw, in-process, and finished drinking water samples using laboratory and on-line TOC analyzers employing the heated sodium persulfate oxidation technique in USEPA-approved methods 415.3⁽⁹⁾ and SM 5310C⁽¹⁰⁾.

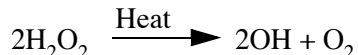
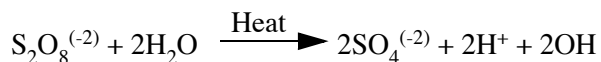
Heated Sodium Persulfate Oxidation

Virtually all organic compounds dissolved in water can be oxidized by heated sodium persulfate ($\text{Na}_2\text{S}_2\text{O}_8$) oxidation. Concentrated solutions (1 or 1.5 M) can effectively oxidize organic matter present in the form of colloids, macromolecules, and suspended solids.⁽¹¹⁾

Sodium persulfate is highly soluble in water:



When heat is applied sulfate and hydroxyl radicals are formed by the following reactions:



Oxidation of organic molecules requires 2.5 - 3 sulfate or hydroxyl radicals per carbon atom.

Experimental

Instrumentation used in this study included an OI Analytical Aurora 1030W laboratory TOC analyzer equipped with a model 1088 rotary autosampler (Figure 1) and a 9210p on-line TOC analyzer equipped with a process gas module to deliver CO_2 -free air (Figure 2). Instrument operating parameters, reagent conditions, and calibration settings used in this study are summarized in Table 1.



Figure 1. Aurora 1030W Laboratory TOC Analyzer with 1088 Rotary Autosampler

The Aurora 1030W and the 9210p TOC analyzers both remove inorganic carbon internally by acidification followed by purging with nitrogen. Complete removal was verified previously by analysis of carbonate solutions. Once inorganic carbon is removed, the persulfate reagent is added and the sample is heated to 98 °C to oxidize and convert all organic carbon to carbon dioxide (CO₂). The CO₂ is quantified by a non-dispersive infrared detector (NDIR) and the result reported as TOC content in both mass and concentration.



Figure 2. 9210p On-line TOC Analyzer.

Table 1. Instrument Operating Conditions

Parameter	9210p	Aurora 1030W
Analysis Mode	NPOC	NPOC
Sample Volume	2 mL	2 mL
Phosphoric Acid Volume	1.0 mL	0.5 mL
Persulfate Volume	1.2 mL	1.2 mL
TIC Reaction Time	0.5 min	1.30 min
TOC Reaction Temperature	98 °C	98 °C
TOC Reaction Time	0.5 min	2.30 min
TOC Detection Time	1.45 min	3.0 min
Calibration Standard	KHP - C ₉ H ₈ O ₄ K	KHP - C ₉ H ₈ O ₄ K
Calibration Points	0, 1, 5, 10, 25 ppmC	0, 1, 5, 10, 25 ppmC
Calibration Blank	0 ppm	0 ppm
Cal 1	1 ppm	1 ppm
Cal 2	5 ppm	5 ppm
Cal 3	10 ppm	10 ppm
Cal 4	25 ppm	25 ppm

Calibration

The same analyst using the same source of neat material prepared reagents for each analyzer. Low TOC laboratory reagent water was used to prepare rinse water, reagents, calibration, and quality control solutions. Certified calibration standards were obtained from Ultra Scientific and are available from OI Analytical. Both instruments were calibrated using Potassium hydrogen phthalate (KHP) as a carbon standard. Carbon-free water was used as a zero standard and to prepare KHP and dilute standards and reagents.

U.S. EPA Method 415.3 specifies the use of at least four calibration concentrations and a calibration blank for a minimum of five total calibration points. SM 5310C specifies an initial instrument calibration using a minimum of three concentrations for linear curves, or a minimum of five for non-linear curves.

Each analyzer was calibrated with four standards plus a calibration blank totaling five calibration points as specified in EPA 415.3. Calibration with multiple standards is required by EPA 415.3 and/or SM 5310C, thus the 9210p on-line TOC analyzer is fully compliant with this EPA method requirement. Each 9210p calibration standard was reanalyzed to verify accuracy across the entire calibration range. Figure 3 plots the actual TOC value of the calibration standards and the calculated value obtained after the standards were re-analyzed and Figure 4 plots the precision of the 9210p at various concentrations.

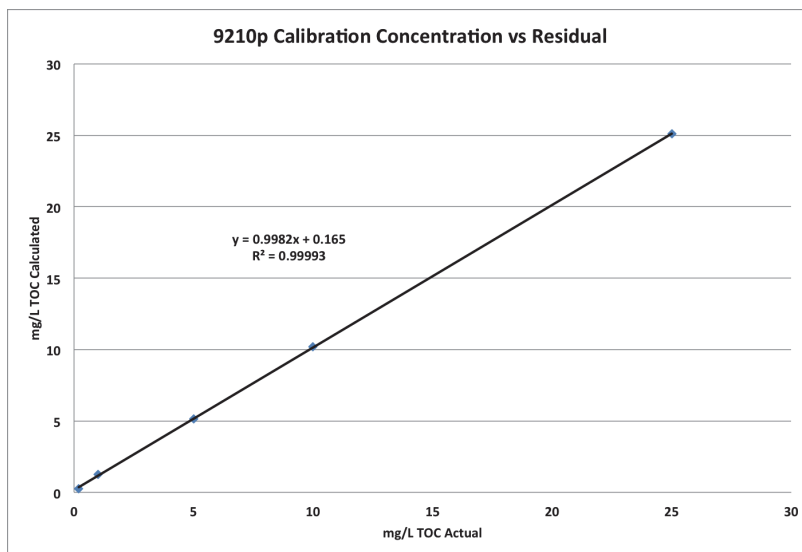


Figure 3. Calibration Accuracy of the 9210p On-line Analyzer

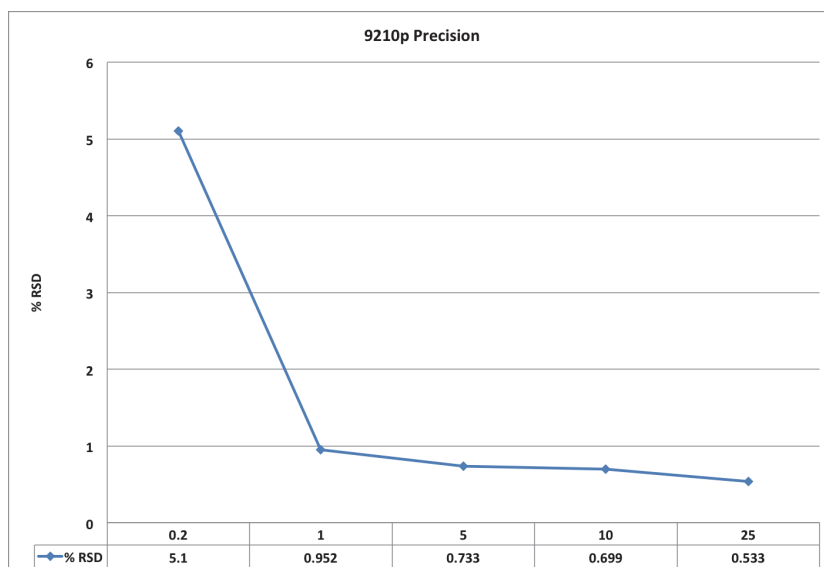


Figure 4. Precision of the 9210p Analyzer

Each instrument was calibrated on the same day using the same calibration solutions. Verification standards and MDL determination was started the day after the instruments were calibrated. The calibration verification standard for the laboratory analyzer did not pass requiring that the laboratory analyzer be re-calibrated before any further analyses. The 9210p On-line Analyzer calibration verification was well within our internal $\pm 10\%$ acceptance limits allowing sample analyses on the on-line analyzer to begin, as the laboratory analyzer was being re-calibrated.

Operating conditions of the analyzers were adjusted to obtain the Safe Drinking Water Act (SDWA) minimum required reporting limit of 0.35 mg/L but also allow for accurate quantitation in storm or terrorist events that may yield TOC values much higher than the average values of 2 - 10 ppm. Calibrated in this way, the 9210 on-line analyzer serves as process control for TOC removal to minimize DBP formation, and also provides an early warning of organic contamination in the drinking water supply system.

Results and Discussion

A PotableWatr™ Quality Control Sample obtained from Environmental Resource Associates (ERA)⁽¹²⁾ was analyzed on each instrument in quadruplicate. Table 2 summarizes the data. Both instruments obtained values very near the certified value of 8.53 mg/L and were well within the 7.21 - 9.85 mg/L acceptable range. Notice that the precision of the 9210p on-line analyzer exceeds that of the 1030W laboratory instrument.

Table 2. TOC Data from Analysis of Potable Watr™ Quality Control Standard

9210p (mg TOC/L)	1030W (mg TOC/L)
8.68 (1.03)	7.92 (0.20)
8.75 (0.82)	8.67 (0.06)
8.76 (0.75)	8.56 (0.63)
8.73 (0.86)	8.41 (3.62)

%RSD in parentheses n=3

The 40 CFR part 136 Appendix B method detection limit was determined on each analyzer using a 0.2 mg/L TOC standard. The 9210p method detection limit (MDL) was 0.05 mg/L and the MDL for the 1030W was 0.07 mg/L. The accuracy of the 9210p and the 1030W MDL standard was 127% and 75% respectively. Each was well within the $\pm 50\%$ EPA 415.3 method requirement. An upper limit of linearity was verified on the 9210p only. Results from a 50-mg/L carbon and a 100-mg/L carbon standard obtained 49.9 mg/L and 98.4 mg/L respectively.

A Suwannee River NOM reference material (1R101N) with known organic carbon content was obtained from the International Humic Substances Society.⁽¹³⁾ The entire portion of humic material supplied was transferred to a 1000-milliliter volumetric flask and diluted to the mark with reagent water to prepare a 52-ppm TOC humic acid quality control standard. This standard was diluted further to 1.04 mg/L TOC and 5.2 mg/L TOC and the solutions were analyzed on each analyzer. Table 3 provides a summary of the results for Suwannee River NOM reference material. The 9210p on-line TOC analyzer obtained better recovery and precision at the higher humic acid concentration.

Table 3. Recovery of Suwannee River NOM Reference Material

9210p (mg TOC/L)	1030W (mg TOC/L)
1.19 (1.15)	1.13 (1.38)
4.55 (0.85)	4.34 (1.55)

%RSD in parentheses n=3

Six surface waters with unknown TOC concentrations were obtained from a local commercial laboratory (Accutest, Bryan, Texas) and analyzed on each instrument. One of the samples was chosen at random as a duplicate resulting in a total of seven samples each analyzed in triplicate. Figure 3 plots the correlation of these unknown samples between the two analyzers.

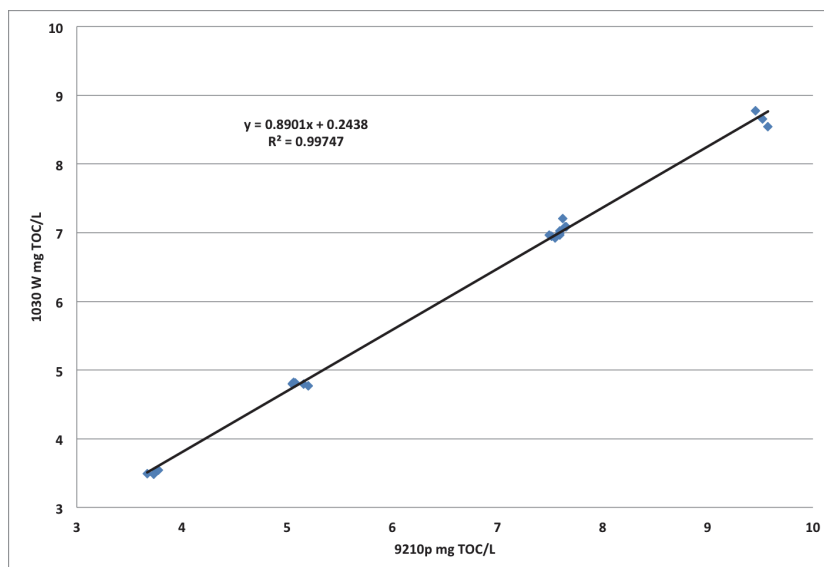


Figure 5. Correlation of Unknown Samples Between the 9210p and the 1030W

The correlation of 0.99747 demonstrates that the results between the two instruments are statistically the same. A T-Test (data not reported here) showed the same. The slope of less than one indicates slightly higher results were obtained with the 9210p on-line analyzer then with the Aurora 1030W laboratory TOC analyzer.

Summary and Conclusions

A 9210p on-line analyzer configured to analyze TOC by heated persulfate oxidation with NDIR detection was evaluated. Results of this study indicate that the 9210p on-line TOC analyzer and Aurora 1030W laboratory TOC analyzer obtain comparable data on drinking water samples.

Unlike most on-line analyzers that calibrate with a maximum of two points, the 9210p allows a 5-point calibration making the analyzer fully compliant with Methods 5310C and 415.3. Comparison data shows that the 9210p obtained results that are equivalent, and perhaps better, than laboratory analysis. The 9210p on-line TOC analyzer is capable of providing continuous unattended TOC data that can be used for routine monitoring of TOC removal and DBP reduction, detecting high TOC events, and drinking water security monitoring. The 9210p is fully compliant with calibration, oxidation, detection, and QC verification criteria required in U.S. EPA approved TOC methods 415.3 and SM 5310C. Accordingly, the results obtained by continuous TOC monitoring should be reportable for compliance purposes.

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